SkinCart 💄

Selfie in. Basket out.

Skin analysis → routine you can preview on your face → foundation shade matched and tried on → one-tap order

Live demo: https://skincart-youcam.vercel.app (no login, "Use sample photo")
Code: https://github.com/heavycastles/skincart-youcam (MIT) · **Video:** https://youtu.be/NIFuCn8lOT0

Problem

Small cosmetics shops in **Central Asia / CIS** sell through Telegram & Instagram DMs.

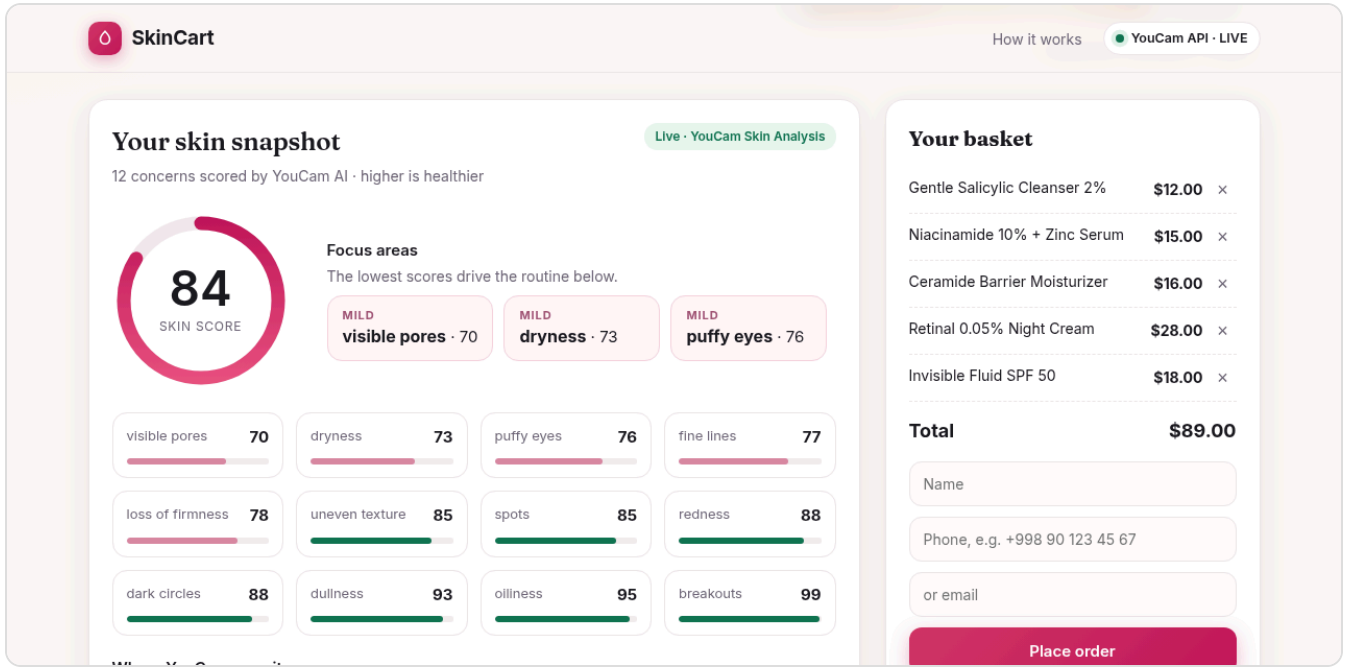
- "What should I buy for my skin?" is answered **by hand**, all day
- Customers **can't test** a foundation or a lipstick in a chat
- The wrong foundation shade is one of the top reasons it gets **sent back**

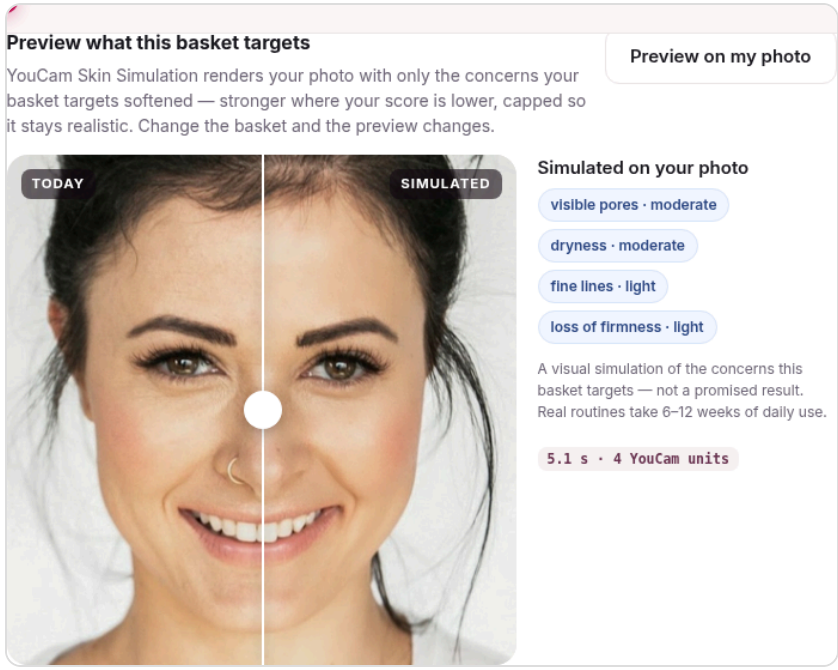
> Give a one-person shop the scan → recommend → try on → buy flow of a big beauty brand.

What SkinCart does — one upload, four YouCam AI APIs

1. **Skin Analysis** scores 12 concerns → top 3 focus areas + YouCam overlay masks
2. Routine from the shop's **own catalog** (always SPF) + **morning / evening plan** with clash rules
3. **Skin Simulation**: the customer's photo with only what the basket targets softened (0.25–0.7 by score)
4. **Facial Color Tones**: undertone + depth → nearest of 12 foundation shades (ΔE), neighbours, finish
5. **Makeup VTO**: the matched foundation and lip shades ranked for the undertone, before/after slider
6. **Order** → scored, de-duplicated lead with skin profile + basket (optional n8n owner alert + day-14 re-scan)

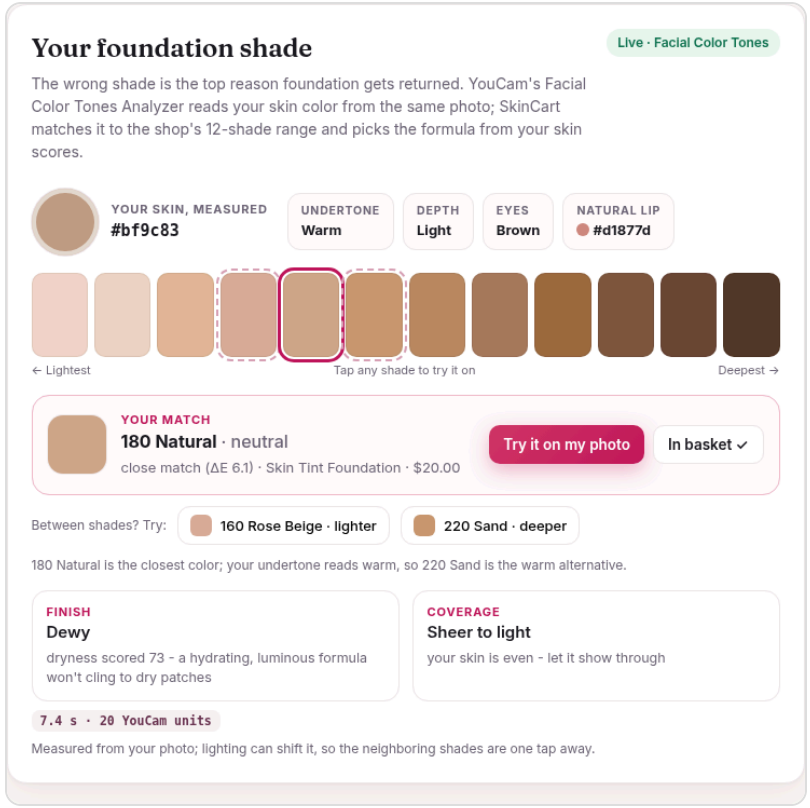
Live: analysis, routine, preview

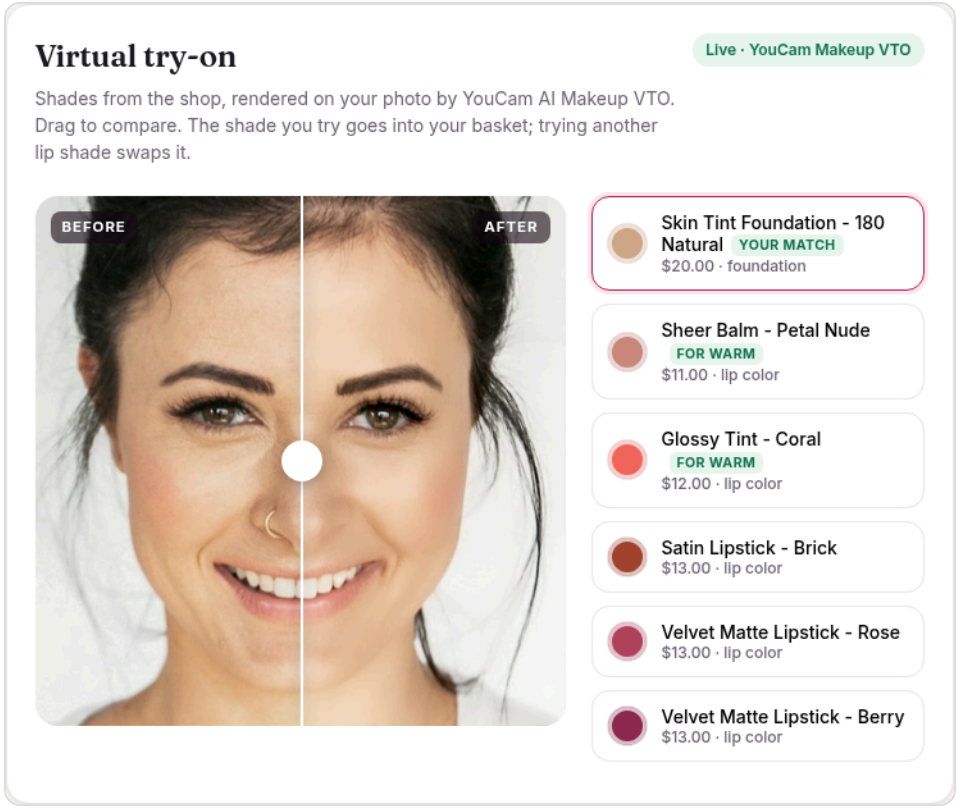




Score 84 · visible pores, dryness, puffy eyes · Skin Simulation of the basket (4 units)

Live: shade match and try-on





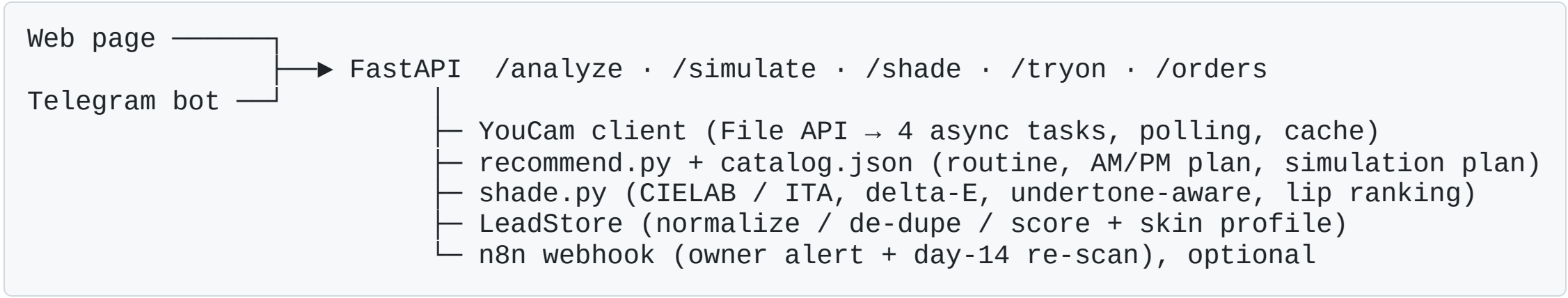
Skin #bf9c83 · warm · light → **180 Natural** (ΔE 6.1) · dewy finish because dryness scored 73

YouCam APIs used

Step	Endpoint	Units
Upload once	`POST /s2s/v2.0/file` → presigned PUT → `file_id`	0
Skin Analysis	`POST /s2s/v2.0/task/skin-analysis` (12 SD actions) → poll	14
Skin Simulation	`POST /s2s/v2.0/task/skin-simulation` (basket concerns) → poll	4–6
Facial Color Tones	`POST /s2s/v2.0/task/skin-tone-analysis` → poll	20
Makeup VTO	`POST /s2s/v2.0/task/makeup-vto` (lip / foundation) → poll	1

Same `file_id` for every task · results cached per photo (repeat = **0 units**) · the page shows units per step

Architecture



Vercel (serverless FastAPI) · Docker Compose · 77 pytest cases incl. headless-browser runs at 1280 and 375 px

Built for real shoppers

- Photo checks **before** any API call (format, size, framing step); YouCam errors become plain advice
- Honest UI: "simulation, not a promise", units shown, mock mode clearly labelled
- Accessible: AA contrast, live announcements, keyboard sliders · Lighthouse 100 on mobile and desktop
- Private by default: contacts never echoed back, lead list is admin-only

Impact & what's next

Impact: a beauty counter for every small shop — the right shade and a realistic preview before buying,
and an order with context instead of "hi, how much?".

Next:

1. CSV catalog onboarding for any shop
2. Uzbek / Russian replies
3. Progress chart from the 14-day re-scans
4. More try-on categories and Instagram DMs

Built with: Python · FastAPI · YouCam AI · JavaScript · n8n · Docker · Vercel · pytest